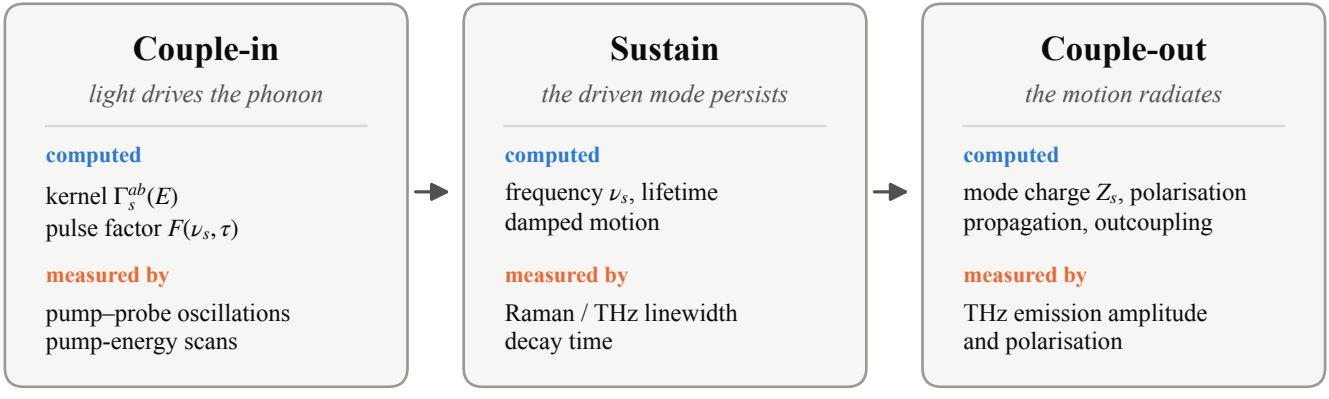

Couple-in
light drives the phonon
computed
kernel $\Gamma_s^{ab}(E)$
pulse factor $F(\nu_s, \tau)$
measured by
pump–probe oscillations
pump-energy scans
Sustain
the driven mode persists
computed
frequency ν_s, lifetime
damped motion
measured by
Raman / THz linewidth
decay time
Couple-out
the motion radiates
computed
mode charge Z_s, polarisation
propagation, outcoupling
measured by
THz emission amplitude
and polarisation
screening observable v2: $T_i^{ab}(E) = \frac{1}{V} \sum_{s \in M} Z_{s,i} \Gamma_s^{ab}(E) \sqrt{\nu_s}$
couple-in to second order; sustain and couple-out idealised